

# Effect size and practical importance

Project: Reproducible AI Research Study

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Uncertainty intervals

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Confounders and controls

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Hypotheses and measurable outcomes

Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Trace the worked case for Effect size and practical importance. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Effect size and practical importance. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Apply a new case independently

Baseline 6, 8, 10; candidate 5, 7, 9. Useful mean gain at least 1.5. Decide.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

### My reasoning and evidence

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## 6. W2 Trace the worked case

Baseline times 10, 12, 8; candidate 8, 9, 7 seconds. Useful gain threshold 2 seconds. Show the intermediate steps and give the conclusion with its scope.

### My reasoning and evidence

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## 7. W3 Audit the limit

A report omits this qualification: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

### My reasoning and evidence

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## 8. W4 Apply a new case independently

Baseline6, 8, 10; candidate5, 7, 9. Useful mean gain at least1.5. Decide.

### My reasoning and evidence

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## 9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

### My reasoning and evidence

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## 10. Project checkpoint and reflection

Add a labelled effect size and practical importance evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of effect size and practical importance. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### My project change, supporting evidence and remaining limit

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## 11. Feedback, return check and learning record

## Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

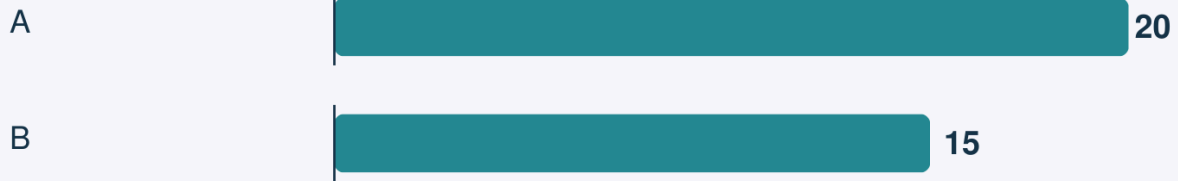
## 12. Pictorial case practice: Effect size and practical importance

An improvement with a practical threshold. A team compares error rates against a predeclared practical threshold.

A makes 20 errors in 1000 tasks; B makes 15. Practical improvement requires at least 10 fewer errors per 1000.

## Errors on 1000 tasks

errors per 1000 tasks



Bar length shows the supplied value; the marked baseline is zero.

## Effect size and practical importance

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Compute rates

A is 2%; B is 1.5%.



#### 2. Compute effect size

B has 5 fewer errors per 1000, a 0.5-percentage-point reduction.



#### 3. Apply practical threshold

Five is less than ten, so the observed change misses this threshold.

### CHANGE ONE THING

B makes 8 errors in 1000.

Explain the changed result using the matching lesson.

Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

## 13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

### Evidence to keep · My record

**Evidence to keep:** This week's input and deciding step

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** Prediction and actual result (or labelled paper trace)

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One change since the previous checkpoint and why

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One error or limit and the next test

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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## 14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

An improvement with a practical threshold. A team compares error rates against a predeclared practical threshold.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Five is less than ten, so the observed change misses this threshold.

1. Apply practical threshold
2. Compute rates
3. Compute effect size

#### My selected conclusion, deciding fact and explanation

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Apply practical threshold Five is less than ten, so the observed change misses this threshold. Effect size describes how much the outcome changes. Practical importance depends on the real decision criterion; relative changes can sound large while absolute changes remain small.

### Practice 2: explain the deciding evidence

Changed-case practice: B makes 8 errors in 1000. Which conclusion is supported by the supplied facts?

1. The reduction becomes 12 per 1000 and meets the stated practical threshold.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

### My selected conclusion, deciding fact and explanation

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The reduction becomes 12 per 1000 and meets the stated practical threshold. Twenty minus eight exceeds the required ten fewer errors.

### Repair a missing building block

Review Uncertainty intervals (ai-7/12). Goal: Explain the mechanism and terminology of uncertainty intervals. Short practice: Locate -0.8 and zero relative to the supplied interval [-1.4,-0.2]. Interpret the sign and confidence-method meaning, then describe what changes with interval [-1.4,0.3].. Return to this week and explain what you changed.

Review Confounders and controls (ai-7/10). Goal: Explain the mechanism and terminology of confounders and controls. Short practice: Draw the four condition/background combinations. Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Review Hypotheses and measurable outcomes (ai-7/7). Goal: Explain the mechanism and terminology of hypotheses and measurable outcomes. Short practice: Name the predicted outcome and how unsupported is scored. If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?. Return to this week and explain what you changed.

### Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

### What I repaired, the evidence I kept and the next test

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